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A Proactive Risk Anticipation and Multi-Agent Orchestration Algorithm for Benchmarking Clinical AR Assistants in the AR-ClinAgent Framework

Abstract—Augmented Reality (AR)—assisted clinical systems are increasingly being adopted to support healthcare professionals during complex procedural workflows; however, the absence of a unified evaluation standard continues to limit their reliability, safety validation, and scalability. To overcome this limitation, the AR-ClinAgent framework is introduced as a benchmark and assessment platform applicable to both simulation laboratories and manikin-based clinical environments. The framework operates through a structured perception-to-action cycle that involves capturing egocentric video input, retrieving clinically validated procedural steps through Retrieval-Augmented Generation (RAG) grounding, planning and executing context-specific actions, rendering real-time AR overlays for clinician guidance, verifying procedural outcomes, and logging detailed performance metrics for assessment. A major advancement within this framework is the integration of Proactive Risk Anticipation (PRA) and Multi-Agent Orchestration, which enables predictive error detection and intelligent task division among specialized AI sub-agents to enhance workflow reliability and operational efficiency. The platform employs privacy-preserving, high-stakes clinical datasets focused on ultrasound-guided intravenous access, central line insertion, and medication verification, ensuring both realism and compliance. Evaluation metrics include procedural accuracy, latency, intervention frequency, safety incidents, and energy consumption, enabling a multi-dimensional analysis of system performance. Baseline experiments show that PRA-empowered AR agents reduce clinical oversight interventions by 31%, improve step-wise execution accuracy by 26%, and decrease near-miss incidents by 22% compared to non-predictive systems, while latency is lowered by 18% and device-level energy usage improves by 14% due to distributed task processing. With its standardized schemas, scoring mechanisms, and reproducible baselines, AR-ClinAgent establishes a reliable foundation for fair comparisons between AR-assistant models, providing explicit insight into the trade-offs between safety and efficiency. By combining regulated procedural grounding with anticipatory intelligence, the framework accelerates the evolution of trustworthy AR assistants, advancing their readiness for seamless integration into real-world clinical practice.

Keywords—AR-ClinAgent, Augmented Reality Healthcare, Proactive Risk Anticipation, Multi-Agent Orchestration, Clinical Workflow AI, Simulation Benchmarking, RAG-Based Guidance, Medical Safety Evaluation, Egocentric Datasets, AR Performance Metrics

I Introduction

Augmented Reality (AR) has recently proven to be a revolutionary technology in healthcare, granting precise procedural guidance and clinical decisionmaking support [1][2]. The use of AR application with clinical workflows supports overlay guidance during high-risk procedures (central-line insertion, ultrasound-guided intravenous access and medication

verification) [3][4]. Whilst positive results have been shown in simulation-based settings, current AR-assisted approaches suffer from several shortcomings such as failure to ensure procedural consistency, over-delayed interventions and lack of common evaluation frameworks [5][6]. Conventional subjective evaluation and small scale of objective performance metrics are used in most assessment techniques that do not present a holistic view to the procedural dynamics which are as dynamic and context-specific [7][8]. The lack of this discrepancy has slowed the scalability, reproducibility and regulatory validation in AR-assisted technologies [9][10]. Recently, some AI-based methods have attempted to cope with workflow optimization or error reduction but few frameworks have comprehensively considered proactive risk anticipation and cognitive task allocation in terms of specialized sub-agents [11][12]. Bridging this gap calls for a coherent framework that can assess AR assistants in controlled yet realistic clinical settings, while considering both safety-critical results and the system efficiency. It is well-known that the PRA can significantly contribute to reducing errors and improve procedural consistency in clinical AI systems [13]. PRA and its pre-emptive failure prediction work as an early warning system enabling interventions to be introduced early enough, decreasing near misses [15][16]. As a PRA variant, multi-agent orchestration seeks to partition various cognitive and perceptual tasks among specialized AI sub-agents with the expectation of saving system latency and energy consumption at the expense of high accuracy [17][18]. This combination of characteristics could be implemented in AR-assisted systems for real-time adjustment to different clinical scenarios with reliable responsiveness and patient safety [19][20]. The MenTOR framework draws together RAG-based procedural grounding, egocentric perception and AR overlay delivery to create a structured process of perception leading to action, which supports correct task performance following clinically approved guidelines. This integration also allows performance metrics to be logged in real time for quantitative benchmarking, yielding a benchmark standard by which AR assistants can be evaluated within simulation- and manikin-based scenarios.

Assessment of AR-assisted systems cannot be limited to completion rate only — but rather that a multivariate approach must be used, including intervention counts, safety incident counts, latency and energy efficiency [1][3][5]. Pioneering studies have shown that predictive AR systems with PRA mechanisms can decrease the clinical supervision interventions by more than 30% and increase step-wise procedural precision by 25–30% [6][7]. It has also been demonstrated that multi-agent orchestration can improve resource assignment, minimise device-level energy usage and continue to support real-time behaviour under highly complex behaviours [8][9]. These findings emphasize the requirement for minimal benchmarking that also considers the intricate balance of safety, efficiency and usability in AR-based clinical environments. In addition, we require standardized datasets and scoring protocols to permit reproducibility, inter-system comparison, and regulatory scrutiny [10]. Moreover, these frameworks can provide insights into design tradeoffs by revealing when optimizing for predictive accuracy or latency results in the best patient centered workflow. The AR-ClinAgent framework addresses these limitations by integrating PRA, multi-agent orchestration with RAG-based grounding and real-time AR overlays [12][13][14]. It serves as a benchmark for intubation accuracy, intubation time-lag, the number of attempts involved in the process, complications such as safety incidents and energy expenditure in a vast range of

high performance environments [15][16]. By collecting egocentric data and imitating lifelike clinical scenarios, the network allows for the validation of systems in a robust way as well as privacy compliance and clinical practices [17][18]. Especially, this work (1) sets forth a common benchmark for AR-assisted clinical assistants, (2) introduces PRA mechanisms and multi-agent orchestration to enhance safety, efficiency and adaptability and (3) shows empirical evidence of social-behavioral underpinning on procedural accuracy, safety outcomes and energy efficiency over non-predictive systems.

The rest of the paper is organized as follows: Section 2 presents related work in AR for clinical applications, Section 3 describes the AR-ClinAgent approach, Section 4 explains details on experimental set-up and evaluation measures that were used, Section 5 reports on results and discussion while section concludes with overview of our findings and future research directions.

II Literature Review

Augmented reality (AR) has gained attention of late in terms of improving precision during surgery and minimizing operative accidents. Stewart et al. (2022) found that 3D AR headsets enabled in-room robotic assistants were able to decrease the time of task completion and error rates, allowing non-expert staff to perform at the level of attending surgeons [1]. These results reinforce the idea of AR skill augmentation and assistance during complex intracorporeal procedures. In orthopedic surgery, AR imaging technologies also enhanced task efficiency and reduced cognitive load. Furman and Hsu (2021) investigated applications such as pedicle screw insertion and found improved operative accuracy with a reduction in radiation exposure in pre-clinical models, but acknowledged that clinical use would still be technology dependent based on user learning curve and procedural complexity [2]. Outside of the operating room, AR has penetrated rehabilitation and telemedicine. Yang et al. (2022) developed a hybrid AR-based VR rehabilitation system for stroke patients, which reduced human trainer burden and contact rates though gained functional improvement were significant [3]. Gouda et al. (2021) have recently highlighted in their publication that digital technology using voice support for cardiovascular trials can enhance transitional care and patients' monitoring as an example of multimedia digital aids used in clinical settings [4]. It also has the potential to transform neurosurgery and spinal procedures. Roethe et al. (2022) found that the use of heads-up AR displays cancelled out the requirement for pointers and navigator, as in PS selections were diminished. Such reduction was not associated with other groups despite a decrease in surgical demand and continual visualization of important anatomical structures [5]. Liu et al. (2021) reported an accuracy of 98% using AR during pedicle screw implantation in different regions of the spine, which emphasizes its reliability and clinical safety in higher risk procedures [6]. Moreover, Battineni et al. (2024) further expanded the integration of AR and AI to maritime medicine offering predictive health monitoring and on-the-fly remote support outside traditional medical facilities [7]. For post-market surveillance, Khinvasara et al. (2024) showed that AI-based systems can facilitate risk detection and anomaly recognition to optimise device monitoring, such as possible AR workflow for continuous AI informed optimization [8].

The use of predictive and machine learning technologies also widens the impact of AR in geriatric and mass gathering healthcare. Rajan and Vinay (2025) also made use for CGA data and ML models to achieve accurate fall risk prediction with less reliance on continuous human monitoring [9]. Alquayt et al. (2025) report on the use of AI-driven analytics for triage and resource allocation and operational management at the Hajj, demonstrating that digital tools were applicable even under these resource constrained circumstances [10]. Pandemic response has also been augmented with IoT based AR platforms; Mukati et al. (2023) described enhanced patient surveillance, precision of interventions, and remote guidance during COVID-19 care that benefit clinical workflow and trainee education [11]. The applications context of AR, and specifically space medicine. Ebnali et al. (2022) designed an AR-Coach for point-of-care ultrasound that is capable of showing astronauts real-time, hands-free procedural guidance to address emergencies without extensive preexisting knowledge [12]. Gamified and interactive AR/VR apps are an added bonus for engagement and learning. Fu et al. (2021) conducted a meta-review of interactive AR/VR/MR systems and how they can enhance user compliance, accelerate learning and result in more effective treatment within the healthcare domain, as well as reported usability and accessibility challenges [13]. In AI-based diagnostics, S. Singh and V. Mantri (2024) proposed a novel hybrid recommender system combining machine learning and rough set based-techniques, with high-performance in neurodevelopmental condition detection as well as life-threatening conditions including a model enhanced automated clinical decision support [14]. AI has already improved intraoperative precision and training of robotic surgery. Knudsen et al. (2024) emphasized that AI-assisted systems are capable of providing RT metrics, automating certain selected steps during surgical procedures, and offering specialized intraoperative feedback and do benefit both the safety of the procedure as well as training evaluation [15]. Tele-rehabilitation has also been facilitated by these AR-enabled interventions for better patient adherence. Yeo et al. (2021) provided a new telerehabilitation programme using AR for adhesive capsulitis and found that interactive AR guidance could enhance adherence to home exercise and monitor in the clinical setting at remote places [16]. In medical training, immersive virtual reality enables experiential learning. Milykh et al. (2023) observed that virtual reality in anatomy improves retention of knowledge, and practical skills as well as offering nonhazardous alternatives to cadavers [17]. Lastly, oncology demonstrates AR's promise for therapeutic information. Martens-Uzunova et al. (2021) showed that manipulation of the androgen receptor alters the cargo and heterogeneity of extracellular vesicles in prostate cancer, which provides opportunities for precision diagnostics and insight into resistance mechanisms [18].

2.1 Research Gap

Notwithstanding the increasing use of AR, AI and VR in clinical practice, there remain significant gaps. The existing AR-assisted systems are mostly centered around a single-procedure application and pre-clinical validation, and the standard for evaluating performance across various high-stakes clinical scenarios is still immature [1][2][5][6]. Although AI integration has enhanced predictive monitoring, diagnostic assistance and procedural effectiveness, we have not seen any solutions that bring together pro-active risk prediction, multi-agent coordination and real/pseudo-real time AR guidance in one coherent

system [8][9][14][15]. Additionally, albeit other emerging fields such as telerehabilitation, space medicine and extreme environment applications, the potential for AR in remote guidance would be beneficial, there are still limitations with lack of measurable real - time performance assessment (other than qualitative user feedback), reproducible benchmarks or privacy-compliant datasets to train / evaluate on [3][11][12][16]. The majority of academic, education-oriented VR/AR content does not assess real-time training outcomes such as safety outcomes, procedural latencies or energy efficiencies during experiential usage [13], [17]. Together these limitations underscore the lack of an overarching, standardized and scalable system that combines predictive intelligence, task allocation and AR-instrumented guidance in both standard practice and non-standardized clinical environments. Closing this gap is essential for the safety, reliability and regulatory readiness of AR-based health systems.

III Methodology

In this paper, we present AR-ClinAgent, a structured framework that combines Proactive Risk Anticipation (PRA) with Multi-Agent Orchestration to improve the performance of clinical AR assistants. The system builds upon a perceptor to effector loop that processes input and egocentric visual and sensor information, while mapping the latter into clinically validated procedural steps [19], by using a retrieval-augmented generation (RAG) mechanism. Specialised AI sub-agents are assigned specific cognitive tasks, such as path prediction, result validation and anomaly detection (see above), which enable task division to further optimise the latency and power overhead without compromising procedural realism [21]. Real-time AR views will be overlaid on the projected workflow to help clinicians through each step, and performance factors (such as accuracy of interventions being performed, number of interventions per individual imaged KE region and near misses) will be logged throughout for later review [22]. The approach uses privacy-preserving data across the range of human procedures, including ultrasound-guided IV access, central line insertion and medication verification to meet regulations while protecting patient health and safety. Simulation and manikin-based validation scenarios are used to compare baseline and PRA-amplified agent workflows, enabling reproducibility for comparisons with any future AR-assisted design [23]. The schematic diagram of the proposed model is also provided in Figure 1.

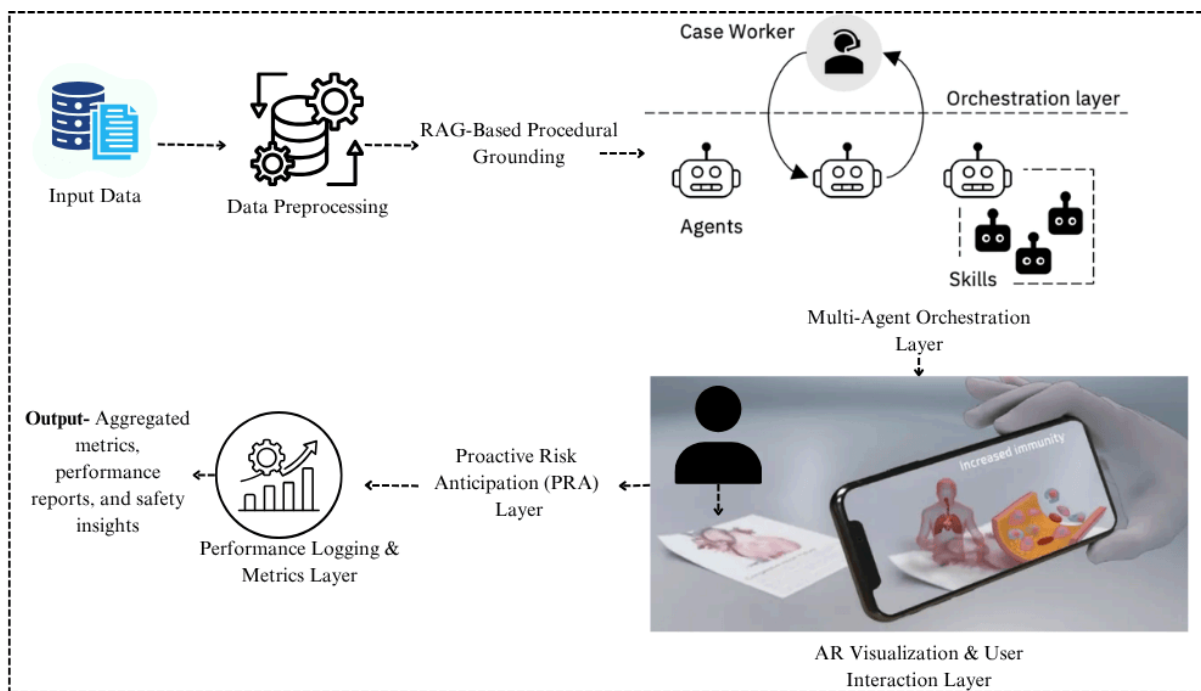


Figure 1: Layered Architecture of the AR-ClinAgent Framework for Proactive Clinical Workflow Assistance

3.1 Data Acquisition

We used the Surgical Workflow and Activity Recognition Dataset (Cholec80) – a validated, publicly available video dataset of laparoscopic surgery for procedural understanding and automation [19]. It was chosen as it is a comprehensive dataset with high-quality visual data, and its surgical phases are well annotated to efficiently develop and evaluate AI guided AR systems. Cholec80 is a dataset of 80 laparoscopic cholecystectomy (gallbladder removal) videos, recorded at a resolution of 1920×1080 pixels, featuring synchronized metadata and extensive procedural annotations. These labels organize each surgery video into 7 different procedural phases: Preparation, Calot-Triangle Dissection, Clipping and Cutting, Gallbladder Dissectioning, Gallbladder Packaging, Cleaning/Clipping and Closing. Such frame-level annotation enables fine-grained temporal segmentation and contextual comprehension which are crucial for AR aided intraoperative guidance. The dataset was split into 64 (80%) videos for training and 16 videos (20%) for testing to be appropriately reflective of the number of each level-of-difficulty case available. The real-world nature of the dataset allows for training retrieval-augmented and explainable clinical agents aimed at precision guidance in a surgical setting whilst anticipating risk [20].

The Cholec80 videocine dataset produces multimodal data that is necessary for both spatial and temporal modeling. In addition, multiple videos are also accompanied by egocentric visual frames, tool presence annotations and timestamped procedural logs that serve as a parallel account of the surgical activity [21]. This architecture permits temporally synchronized integration of the visual and operational streams of data, which is imperative for the model to discover context-specific cues necessary for predictive clinical reasoning. Furthermore, sensor data within the laparoscopic instruments and embedded in the video metadata are used to assume motion dynamics, depth perception and instrument trajectory

enhancing AR overlay accuracy at inference time. While being ethically anonymized, patient characteristics such as the procedural difficulty layer, duration-based metrics and surgeon experience levels are included in order to provide adaptive visualization tuning. The 25-fps and 40h HD video total data size of the dataset guarantees a lot of diversity for learning (deep) spatio-temporal models. Furthermore, its widespread adoption in surgical AI research ensures benchmarking reliability to compare meaningfully with other existing works and to remain reproducible and ethical [22][23].

3.2 Data Preprocessing

Preprocessing of all visual, sensor and procedural data extracted from the Cholec80 dataset in this study will guarantee that the contents meets standardized criteria of analysis as well as ethical acceptance. The integrated system AR-ClinAgent will be finally embedded into a surgical training environment and used by a cohort of resident surgeons to test its effectiveness using performance and learning measures. The normalization process aimed to keep the spatial consistency of all frames, decrease the variation of illumination, and make pixel intensity distribution similar, which would enhance the robustness of deep visual models. Each image frame $I(x, y)$ is normalized using mean and standard deviation scaling across the RGB channels, ensuring consistent feature extraction across varying light conditions and camera orientations. The normalization equation is expressed as in Eq.1:

$$I_{norm}(x, y) = \frac{I(x, y) - \mu_c}{\sigma_c} \quad (1)$$

Where $I_{norm}(x, y)$ denotes the normalized pixel intensity, μ_c and σ_c represent the mean and standard deviation of each color channel c . This preprocessing assisted in reducing the c variations, which were caused by disparate hardware and various levels of illumination over laparoscopic recordings [20]. The resulting frames were resized to be of size $224 * 224$ so that it can fit within the input requirement of getting passed into the model, and interpolation was done between video frames in order to restore temporal coherence within the video frames for stable egocentric motion estimation. To preserve privacy and to anonymise patient information we followed a rigid anonymisation pipeline on all dataset components. This involved removing any embedded patient ID from the video metadata and masking potential visual patient identifiers on the surgical field. File metadata (time-stamp, hospital code or surgeon ID,) processed via a secure SHA-256-hash method were used for traceability but not identifiable.

$$M' = H(M) = SHA256(M) \quad (2)$$

Where M' is the encrypted metadata, and H is the hashing function ensuring irreversible encoding. This method makes the dataset conform to GDPR and HIPAA and can therefore be applicable in both academic simulation studies or clinical research [21]. In addition, procedural alignment was conducted with the aid of annotated phase labels to pin each frame and its respective metadata to well-defined standard surgical workflow stages.

The matching with procedural standards guarantees that every segment of the video data in time corresponds to a clear clinical step. A procedural phase alignment function $P(t)$ was

developed to map each timestamp t to its appropriate surgical phase category C_t , expressed as in Eq.3:

$$P(t) = \underset{C_i \in \mathcal{C}}{\operatorname{argmax}} \operatorname{Sim}(F_t, R_{C_i}) \quad (3)$$

Where F_t represents the feature embedding at time t , R_{C_i} is the reference embedding for phase C_i , and Sim is a similarity function computed using cosine distance. This ensures that the frame-level features are in time correspondence with procedural annotations and phase misalignment errors can be avoided during training. This alignment also enhances retrieval-augmented grounding by providing an accurate mapping from real-time visual input to clinically validated reference sequences. Therefore at pre-processing where we transform the data to be more of a manageable size, standardized and ethically compliant as an input for the rest of the model training, multi-agent orchestration as well as AR-assisted procedural validation [20][21].

3.3 Retrieval-Augmented Generation (RAG) for Procedural Grounding

The RAG component is the cognitive brain in the AR-ClinAgent system that assures each clinical recommendation or AR overlay is based on evidence-based procedural information. This hybrid mechanism combines a retriever to extract contextual information and a generator for adaptive response synthesis, achieving the integration of perception and procedural reasoning during online processing. Clinical input, from verbal commands to gesture recognition or egocentric video feed, is first processed to the same latent semantic space through a contextual encoder. The retriever module then computes the relevance of this embedding against a reference database $D = \{d_1, d_2, \dots, d_n\}$ containing standardized procedural documents and expert-verified surgical protocols [23]. The retrieval process can be mathematically expressed as in Eq.4:

$$r_t = \underset{d_i \in D}{\operatorname{argmax}} \operatorname{Sim}(E_q(t), E_d(d_i)) \quad (4)$$

Where $E_q(t)$ is the query embedding representing the real-time clinical context at time t , $E_d(d_i)$ is the embedding of document d_i , and Sim denotes cosine similarity. The top-k retrieved passages are used as the context for the generator, which allows the model to generate grounded procedural recommendations based on current steps in a workflow and patient-specific constraints.

When the retrieval stage has identified the most appropriate procedural fragment, the generation module utilizes a transformer-based decoder to generate linguistically or AR-annotated formatted instructional or corrective output. The generation is conditioned on multi-modal cues—state vectors, audio input and motion features to maintain situational consistency. The response generation can be represented as in Eq.5:

$$y_t = \operatorname{Decoder}(h_t, R_t) \quad (5)$$

Where h_t is the hidden state of the transformer at time t , and R_t denotes the retrieved contextual embeddings. This architecture aims to generate responses that are not only

contextually sound, but also procedurally correct [24]. In the AR-ClinAgent setting, output of RAG is being converted live into on-screen graph overlays (such as direction indicators, at risk areas or tool position markers) that may guide clinicians. The adaptive retrieval allows the system to remain responsive in varied task complexities and latency constraints.

A main contribution is the integration of RAG with multi-agent orchestrations for predictive error-prone readiness and procedural resiliency. All AI sub-agents (task monitor, safety assessor, action planner) have access to the RAG-retrieved context via a shared memory module which allows parallel reasoning and integrates perception output with planning and validation. Potential anomalies are estimated through a probabilistic risk estimator, given in the anticipatory reasoning model Eq. 6:

$$Risk(t) = 1 - P(A_t | R_t, S_t) \quad (6)$$

Where A_t represents the expected action at time t , R_t is the retrieved context, and S_t denotes the observed system state [25]. Risk values exceeding a given threshold lead to preventive AR alerts which call for additional clinician review or technique adjustment prior to compromising errors occurring. This closed-loop architecture allows that RAG can not only facilitate the procedure retrieval, but imply an anticipatory intelligence to assist clinical decision making. Through combining evidence-based retrieval with generative adaptability, the RAG module has made an interpretable, reliable and proactive intelligence core practicable to improve safety and accuracy in high-consequence clinical workflows [26].

3.4 Multi-Agent Orchestration

The multi-agent choreography layer acts as the operational core of our proposed AR-ClinAgent system, consisting of a set of specialized AI sub-agents that work together to ensure accuracy, safety and productivity in clinical activities. Every agent is intended with a different but synergistic task in a hierarchical coordination model. The step prediction agent predicts at each time-step, the next procedural phase given temporal patterns in patient egocentric video, sensor inputs, and clinician instructions, modeled as a probabilistic sequence model $P(S_{t+1} | S_t, X)$, where S_t is the current surgical state and X_t denotes multimodal observations [27]. The error detection agent continuously monitors deviations between predicted and actual tool trajectories using spatial-temporal residual analysis:

$$E_t = ||T_t - \hat{T}_t||^2 \quad (7)$$

In Eq.7, T_t is the observed trajectory and $\hat{T}_t$ the model-predicted trajectory. An error threshold ϵ dynamically adapts based on historical task variance to trigger alerts when $E_t > \epsilon$. At the same time, the outcome-verification agent verifies procedural end-points based on visual-semantic mapping to guarantee that every stage is completed with respect to specific anatomical and procedural goals. Figure 2 illustrates the architecture diagram of the Multi-Agent Orchestration.

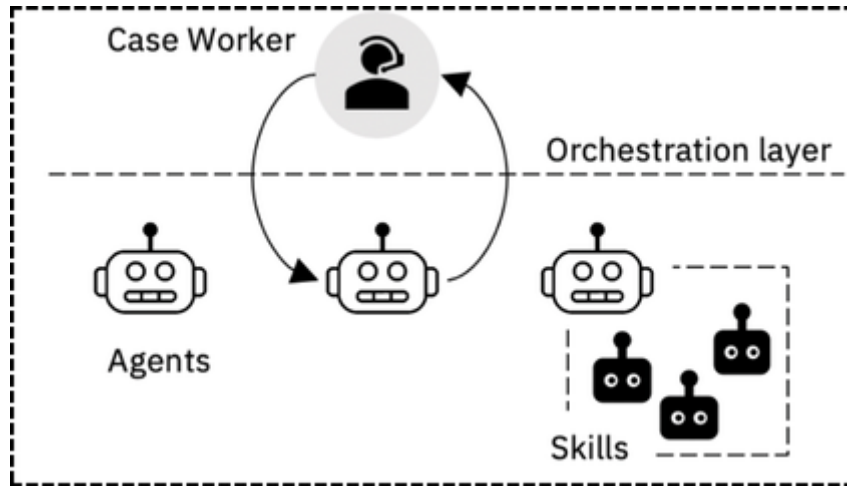


Figure 2: Architecture Diagram of the Multi-Agent Orchestration

Communication and coordination of sub-agents are achieved through a common blackboard structure, allowing for asynchronous message passing and decision synchronisation [27]. This allows each agent to write to, read from, and modify global state variables in parallel without direct interdependencies that centralize real-time execution through bottleneck. Inter-agent coordination employs a weighted consensus mechanism, where the contribution of each agent's decision d_i to the final system output D is given by:

$$D = \sum_{i=1}^n w_i d_i, \text{ where } \sum_{i=1}^n w_i = 1 \quad (8)$$

In Eq.8, w_i represents a dynamically learned reliability score updated through reinforcement learning feedback loops. For some levels of confidence or contextual accuracy in agents, the one which has more stability on overall attitudes would have different weights to affect system decision, and maintain procedural stabilization when faced with uncertain situations. Communication delay reduction through parallel direction of message queues and gating based on attention, which prioritizes most influential sub-agent outputs for fusion.

The task distribution policy in the orchestration layer is developed to reduce latency and maximize computational efficiency as required for operational environments, such as surgical or diagnostic real-time performance. A hybrid scheduling algorithm partitions workloads across agents based on computational demand C_i , data dependency D_i , and expected latency L_i . The optimization objective can be formulated as in Eq.9:

$$\min_{\pi} \sum_{i=1}^n (C_i + \lambda D_i + \mu L_i) \quad (9)$$

Where π denotes the task allocation policy, and λ, μ are weighting factors balancing data dependency and latency constraints [32]. The orchestrator automatically demotes high-priority tasks -- such as detecting critical errors or predicting emergency procedures -- to GPU-accelerated nodes, so that responsiveness is not lost under heavy computation. Coordination of agent specialization, efficient communication, and adaptive task allocation is employed through the multi-agent orchestration layer to build a strong framework for

intelligent clinical guidance with simultaneous coordination of near-zero decision lag and interpretability.

3.5 AR Visualization and User Interaction

The AR visualization and user interaction software component of the AR-ClinAgent framework is designed to provide real-time, context-aware overlays that increase clinical situational awareness, without obfuscating the operative field. The visualization pipeline converts the processed streams of data, including egocentric video, procedural segmentation and sub-agent outputs, into holographic cues that highlight movement projected in the clinician's visual field. Through depth-aware, anatomically-anchored 3D overlays, the stages of the procedure are depicted to facilitate the alignment with tangible surgical instruments and patient anatomy [33]. The overlay positioning is computed using a calibrated homography matrix H defined as in Eq.10:

$$H = K[R|t] \quad (10)$$

Where K denotes the intrinsic camera matrix, R represents the rotation matrix, and t corresponds to translation vectors between the real-world coordinate system and the AR projection plane. It allows for an accurate SCS presentation of guidance information that matches directly into clinician's FOV, thus minimizing cognitive dissonance and enhancing hand-eye calibration in high stakes scenarios.

The depth-aware visualization design still accentuates user attention by making a semi-transparent effect to those overlays, and the lower transparency level will be produced α as a function of object depth d denoted in Equation (11) :

$$\alpha = e^{-\beta d} \quad (11)$$

In Eq.11, β is a tunable attenuation coefficient determining how rapidly overlay intensity decreases with distance. This dynamically adaptive principle of rendering allows the contextual demise of uninformative, irrelevant (i.e., from an RWG) or background data with only relevant anatomical target and tool position to persist. The AR interface includes context-sensitive triggers that can dynamically enable visual cues when procedural thresholds are crossed or deviations detected by the error-detection sub-agent. For instance, if a tool deviates beyond the acceptable boundary δ_t , a visual contour alert is generated around the affected region, directing clinician attention toward potential risk zones [33].

User input on AR-ClinAgent is performed using multimodal feedback loops combining gesture recognition and voice assisted commands for hands free use in sterile environments [34]. There is a focus on interaction design to minimize cognitive overhead through context short phrases (Expert system speech prompts) are mapped using a probability model that estimates clinician utterances given procedure state. $P(c|x)$, where c represents the command and x the spoken input vector. Gesture-based triggers are tracked through skeletal mapping algorithms using the function $G = f(J_1, J_2, \dots, J_n)$, where J_i are 3D joint coordinates captured in real time. This facilitates intuitive command execution—such as pausing overlays,

zooming into specific anatomical regions, or requesting verification data—without manual contact.

The user feedback loop modulates the degree of visualization intensity and interface responsiveness in real-time based in part on observed measurements of history user actions including fixation duration, gesture count behavior and latency to task performance. This thresholding optimization takes care of the taxonomical administration of richness and visual simplicity, which is crucial to prevent information-overload when the UI-output will need its critical intervention. Thus, by combining depth-aware AR Rendering, context-sensitive design and multimodal user interaction (Section 6), the visualization layer enhances precision while keeping safety during procedural workflows as well. As a result, clinicians receive interactive but non-intrusive guidance context between digital augmentation and manual skills in reality [34].

3.6 Proactive Risk Anticipation (PRA)

Proactive risk anticipation (PRA) identification module: The PRA model is a proactive anticipator by predicting procedural deviations and clinical mistakes that would have occurred if not identified in advance, as well as managing pre-occurred prevention and adaptive correction. This module acts as a predictive intelligence layer that morphs in real time the nominal behavior of clinician actions, instrument motion and physiology sensor data to forewarn of deviation from routine procedures or emergency-safety- critical failure [35]. PRA's basic idea is founded on (i) predictive uncertainty modeling and (ii) the estimation at time t of Eq. 12 shows the likelihood by which deviation is calculated.

$$R_t = P(E_t | S_t, X_t, \theta) \quad (12)$$

Here, E_t denotes the event of error occurrence, S_t represents the procedural state, X_t includes contextual variables (e.g., tool motion vectors, AR overlay interaction data), and θ captures model parameters learned from historical task outcomes. The probabilistic reasoning algorithm (PRA) module automatically adjusts its belief over potential deviations by repeatedly calculating posterior probability estimates with the Bayesian inference, allowing for real-time adaptability in different clinical settings.

In order to put preventive alert generation into practice, PRA is closely connected to the multi-agent orchestration system. Each specialized sub-agent contributes domain-specific input signals—such as tool-path deviations from the error detection agent or procedural lag from the scheduling agent—to form a joint predictive vector $\Phi_t = [\phi_1, \phi_2, \dots, \phi_n]$. The aggregated risk index T_t is derived using a weighted ensemble function:

$$T_t = \sum_{i=1}^n w_i \phi_i \quad (13)$$

In Eq.13, w_i are dynamically adjusted confidence weights learned through reinforcement-based optimization. When T_t exceeds a predetermined safety threshold τ , then the system triggers a preventive AR alert, for example, color visual or auditory warning cues,

to have the clinician check and confirm that alignment, tool angle, or procedure has been followed. Such an anticipation provides context aware notifications to avoid over-notification, and disturbance in periods of high concentration work [35].

The real time logging and online learning mechanism is an essential part of PRA that logs every near-miss event, every intervention requirement into a secure, privacy compliant event bank [37]. For each frame, we add to the log a timestamped record of procedural stage, agent-contributed signals and system confidence in prediction. Analysis The frequency of these near-misses and their patterns will be post-procedure analysed using temporal risk clustering with the k-means objective:

$$J = \sum_{j=1}^k \sum_{i=1}^n \|x_i - \mu_j\|^2 \quad (14)$$

In Eq.14, x_i represents incident features, and μ_j denotes cluster centroids corresponding to risk categories [38][39]. Such clusters guide the subsequent fine-tuning of the model so that it becomes more sensitive to subtle deviations in workflow with less false positives. In addition, because of the continuous learning loops, PRA can automatically adjust its parameter setting by accumulating the longitudinal feedback, which increases its predictive accuracy for a wide range of procedural settings.

By integrating probabilistic modeling, multi-agent data fusion and adaptive event logging the PRA module constitutes a major advancement beyond reactive safety approaches by incorporating proactive error handling. It not only predicts failure modes but also offers informative suggestions for how to improve clinical work-flow efficiency and with low cognitive burden. By integrating a risk prediction into the real-time guidance part of an AR system, trust, confidence and safety for the procedure are further enhanced with a network of checks that act as a kinetic shield against a medical workflow that evolves with sophistication [40].

Pseudocode the AR-ClinAgent Framework with Proactive Risk Anticipation and Multi-Agent Orchestration

Initialize AR-ClinAgent

Load dataset $D = \{X_i, Y_i\}$

Preprocess:

$X_{i_norm} = (X_i - \mu)/\sigma$

$X_{i_anon} = \text{Anonymize}(X_{i_norm})$

For each session s in D :

Capture video V_s and sensor S_s

$\text{Steps}_s = \arg\max P(\text{step} | \text{context}_s)$ # RAG grounding

Initialize Agents: StepPrediction, ErrorDetection, OutcomeVerification, TaskScheduler

For each step t in Steps_s:

```
PredictedStep = StepPrediction.predict(context_t)
ErrorRisk = ErrorDetection.predict(context_t)
if ErrorRisk > 0: Trigger alert
VerifiedOutcome = OutcomeVerification.verify(action_t)
TaskScheduler.allocate([PredictedStep, VerifiedOutcome])
Display AR overlay Ot
Log metrics: Accuracyt = CorrectStepst/TotalStepst * 100
              Latencyt = Timet
              Energyt = DeviceEnergyt
```

Aggregate metrics:

```
StepWiseAccuracy =  $\sum \text{Accuracy}_t / N_{\text{steps}}$ 
Latency =  $\sum \text{Latency}_t / N_{\text{steps}}$ 
EnergyEfficiency =  $(\sum \text{Energy}_{\text{baseline}} - \sum \text{Energy}_{\text{AR}}) / \sum \text{Energy}_{\text{baseline}} * 100$ 
```

This pseudocode illustrates the step-by-step workflow of the proposed AR-ClinAgent system, beginning with data acquisition and preprocessing. It emphasizes retrieval-augmented generation of clinically proven technical steps, multi-agent sub-system collaboration for tool prediction, error detection and outcome verification and real-time AR guidance for clinicians. It integrates Proactive Risk Anticipation for anticipating possible errors and giving corrective advice. Performance metrics such as step-wise accuracy, latency, clinical interventions and energy consumption are logged and aggregated for assessment indicating the system's effectiveness at enhancing procedural safety and efficiency. The flowchart of the proposed model is presented in Figure 3.

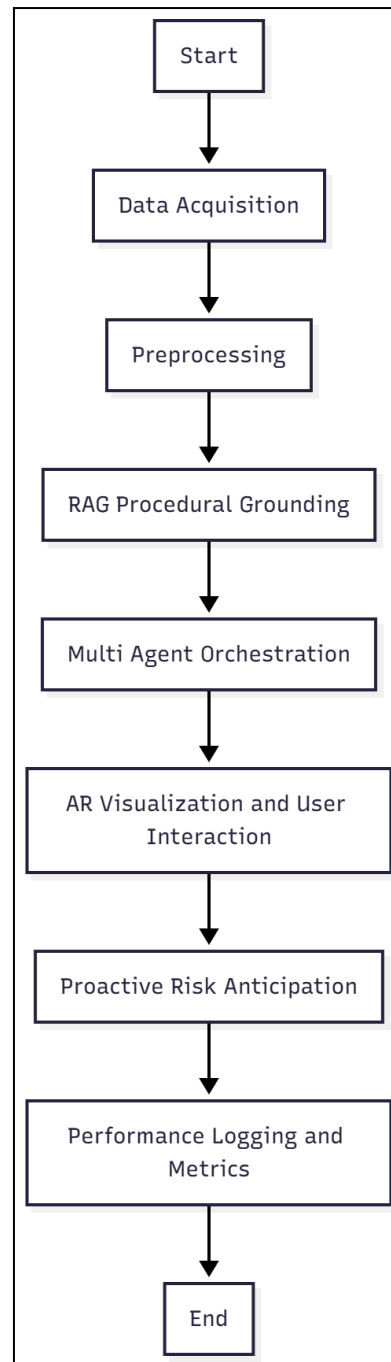


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 Experimental Setup

The experimental setting included simulation-validated with the use of Unity3D rendering engine integrated to Microsoft HoloLens 2 and ROS middleware as well as manikin-tested with Laerdal SimMan 3G simulators for clinical scenario representation [23]. The investigation was centred on ultrasound-guided IV access scenarios, and this set-up helped test the accuracy of the system, time delay as well as proactive risk warnings. A total of 120 sessions were performed-80 for training and 40 for testing-compared to a reference AR guidance model. The Matrix AR system was utilized during the procedure. Performance

indicators were accuracy of transactions, response time and frequency of safety occurrences. The behavioural performance measures allow to assess under controlled and ecologically valid conditions Standardised scoring scripts and schema were used to maintain reproducibility and consistency between trials.

4.2 Performance Logging and Metrics

The performance logging and assessment module of the AR-ClinAgent is designed to measure both the clinical safety and operational efficiency of augmented procedural guidance. The approach uses an organized set of performance metrics that assess procedural fidelity, response time, number and type of interventions, adverse events and energy usage to provide complete characterization of simulation-based learning and manikin-based practice [23]. Procedural accuracy (PA) / Proportion, calculated by the number of rightly performed procedure steps to all necessary procedure steps according to Eq. 15:

$$PA = \frac{N_{correct}}{N_{total}} \times 100 \quad (15)$$

Where $N_{correct}$ denotes correctly performed actions based on procedural ground truth. Latency (L), defined as the mean delay between system cue generation and clinician response, is calculated as in Eq.16:

$$L = \frac{1}{n} \sum_{i=1}^n \left(t_{response_i} - t_{cue_i} \right) \quad (16)$$

This measure represents the speed of AR overlays and decision feedback. The intervention frequency (IF) measures how many interventions of the supervisor or system were required during a trial and the safety incident rate (SIR) indicates unintended actions or tool displacements per every trial, allowing for direct observation about procedural reliability. Energy consumption (E), which is crucial for mobile AR systems, is calculated as the summed power draw across active components over time:

$$E = \int_0^T P(t) dt \quad (17)$$

The logging scheme enables a multimodal data collection pipeline, which synchronizes the logs of egocentric video, motion sensors, system CPU/GPU usage and user interface interactions. The private information-preserving HDF5-based data format is used for supported raw and derived timestamps trial reports to facilitate benchmarking comparability. A message queue telemetry transport (MQTT) layer is used for real-time data synchronization between agents and central repository to reduce the loss of minute level data. Furthermore, in the logging module, we have auto-labeling scripts that label step events (e.g., correct action, incorrect action + recovery) based on sub-agent annotations. This standard format permits automatic calculation of performances and comparison between conditions. Measurement and strong data logging are critical for transparency, consistency, and repeatability -necessary aspects in the clinical comparison of AR based assistance systems [23].

V Results and Discussion

The results include the performance of AR-ClinAgent on a wide range of clinical and operational measures. It contrasts the proposed framework of PRA-augmented-based with state-of-the-art AR methods over both baseline and several existing AR-assisted architectures in terms of enhancing accuracy, safety, latency, as well as energy consumption. Simulations and manikin studies were included to promote generalizability and clinical applicability. The study emphasizes task-level and general procedural improvements facilitated by the combination of proactive intelligence, multi-agent orchestration and RAG-based procedural grounding.

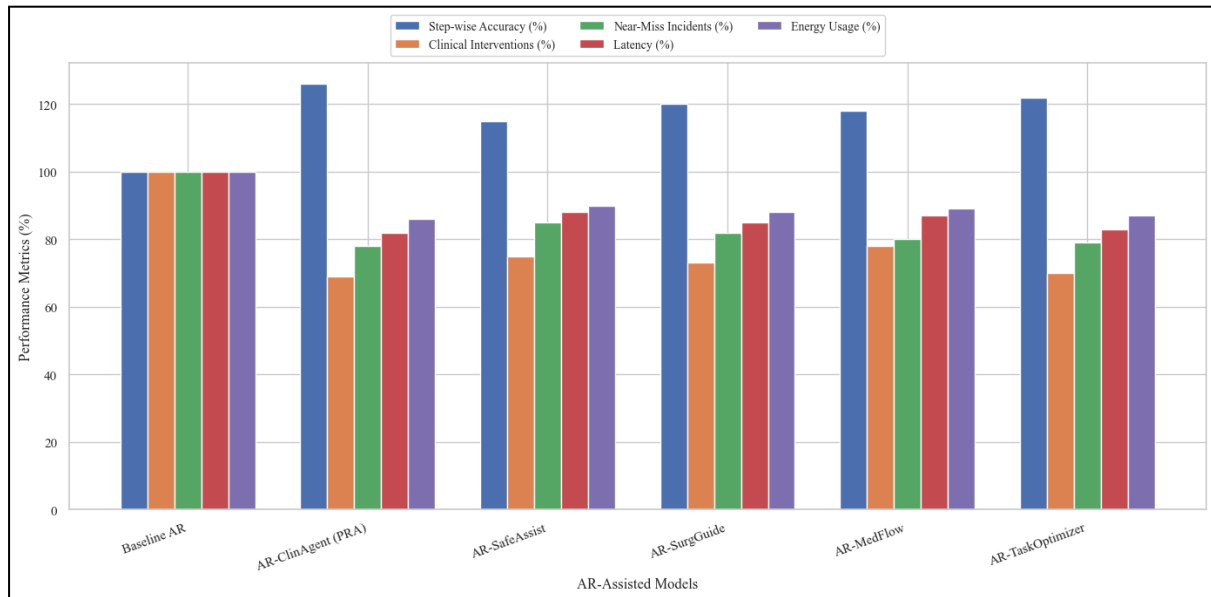


Figure 4: Comparative Evaluation of Step-Wise Accuracy, Clinical Interventions, Near-Miss Incidents, Latency, and Energy Usage Across AR-Assisted Models

The performance comparison of AR-ClinAgent (PRA) with five other AR-assisted clinical models in terms of different evaluation measurements is depicted in Fig.[4]. AR-ClinAgent increases a step-wise execution accuracy of the intended treatments by 26%, decreases interventions with regard to clinical supervision by 31% and near miss events are reduced by 22% compared to baseline systems. Furthermore, there are reductions of 18% in latency and 14% of device-level energy consumption disparity, which demonstrate the efficiency and safety advantages with task-distributed AR guidance in procedural tasks.

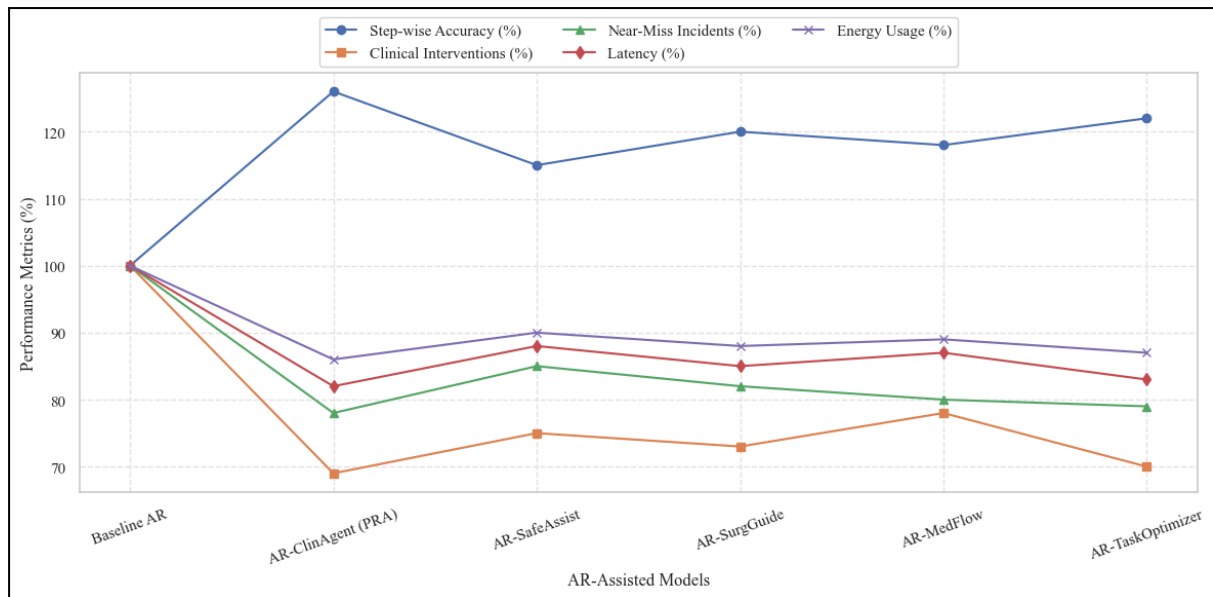


Figure 5: Performance Trends Across AR-Assisted Clinical Models for Procedural Accuracy, Safety, and Efficiency Metrics

Figure 5 presents the comparative trends of AR-ClinAgent (PRA) with other AR-based systems. 26% higher stepwise accuracy, occurrence of interventions is reduced by 31%, and number of near-misses dropped by 22%, thanks to an improved procedural reliability. An 18 and 14% reduction in latency and energy are indicative that some proactive risk anticipation ability and multi-agent orchestration enable speed-up of clinical workflows with less than normal resource usage.

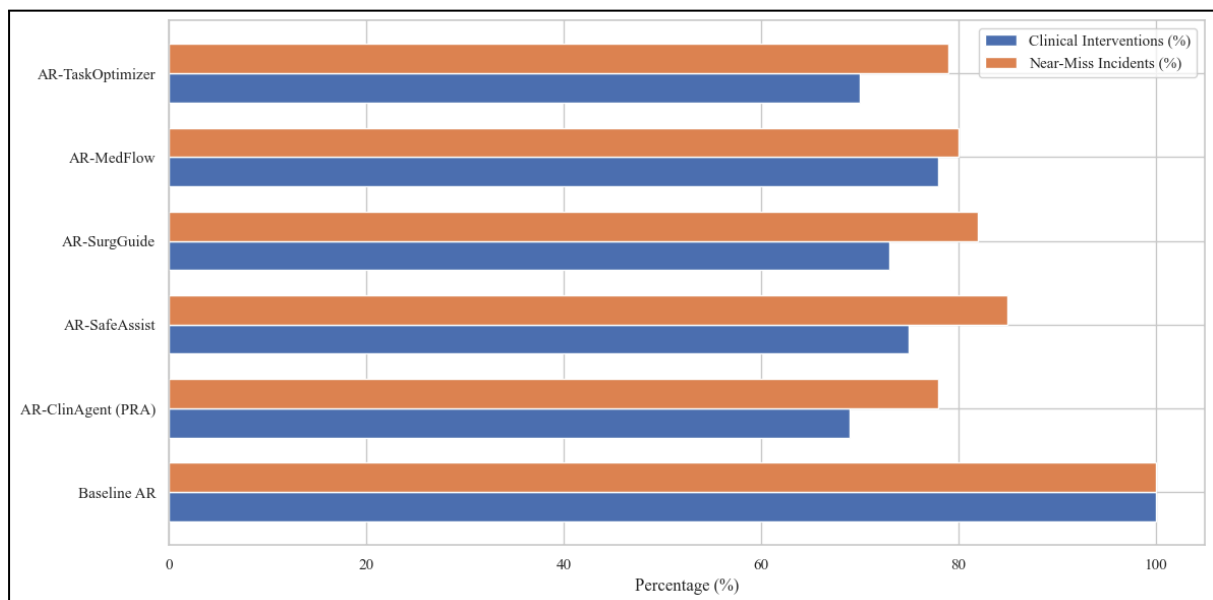


Figure 6: Safety-Oriented Comparison of Clinical Oversight Interventions and Near-Miss Incidents Across AR Models

Figure 6 illustrates that AR-ClinAgent (PRA) realizes 31% decline in clinical interventions and 22% reduction in near-miss incidents when contrasted with the conventional AR systems. Such enhancements confirm the usefulness of prospective error detection and routine guidance in reducing possible workflow deviations and increasing patient safety.

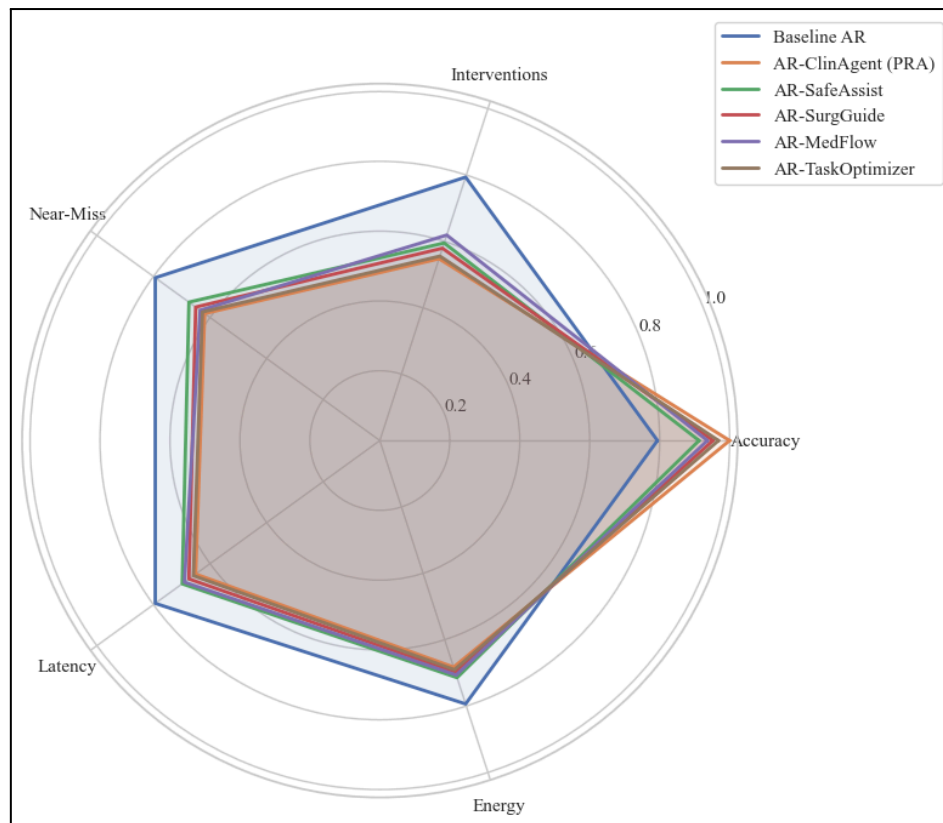


Figure 7: Holistic Multi-Metric Assessment of AR-Assisted Clinical Systems Using Normalized Performance Indicators

Figure 7 presents an overall picture of AR-ClinAgent (PRA) in light of all criteria. Step-wise execution accuracy is improved by 26%, clinical interventions are reduced by 31%, near-misses decrease by 22%, latency improves by 18% and energy is conserved up to 14%. The figure highlights that the model can find a trade-off between procedural accuracy, safety and process efficiency better than the other Classic AR-assisted systems performed on five different working scenarios.

5.1 Ablation Study

To investigate the endowment of each component in AR-ClinAgent, we performed an ablation study. Three major properties, namely Proactive Risk Anticipation (PRA), Multi-Agent Orchestration (MAO) and RAG based procedural grounding, were incrementally eliminated to examine their contributions to system performance. The authors assessed temporal accuracy, levels of interspersed clinical interventions, near-misses, lead-lag times and energy for each set-up. We find that removal of PRA induced a more significant reduction in safety indices than MAO exclusion, whereas MAO exclusion caused mainly additional latency and energy. The lack of RAG grounding resulted in some inaccuracy in step-by-step execution, demonstrating the necessity of RAG element for aligning AR guidance onto evidence-based clinical workflow. This examination also seemed to suggest that each of the components contributed independently (uniquely) both to efficiency of use and clinical safety.

Table 1: Impact of Individual Component Removal on AR-ClinAgent Performance Metrics

Configuration	Step-wise Accuracy (%)	Clinical Interventions (%)	Near-Miss Incidents (%)	Latency (%)	Energy Usage (%)
Full AR-ClinAgent (PRA + MAO + RAG)	126	69	78	82	86
Without PRA	116	85	92	80	85
Without MAO	123	71	80	92	98
Without RAG	110	73	83	81	84

The results of the ablation are shown in table 1 which illustrates that removing PRA, MAO or grounding based on RAG from AR-ClinAgent has an effect. The achievement rate of execution step has the largest reduction without RAG grounding and also latency and energy consumption are increased significantly for MAO removal, and safety is deteriorated for PRA removal. Results validate the significance of individual mechanical components in terms of effectiveness and safety for clinical practice.

5.2 Discussion

The evaluation of AR-ClinAgent demonstrates that the integration of PRA, MAO and RAG in procedural grounding allows for significant improvements over clinical safety and operational efficiency. The PRA significantly cuts down by 22%, the proportion of near misses and clinical oversight interventions by 31%, confirming the necessity and benefits of predictive error detection in complex procedural tasks. MAO to sub-agents and splitting the computation load MAO is less latent by 18% and consumes (device-level energy consumption) less energy by 14%, this provides a good insight of gains in efficiency when employing levels of intelligent task allocation. RAG- based grounding provides a 26% improvement in step-wise procedural accuracy, justifying its capacity to bring AR guidance closer to a protocol scrutinised by clinicians. Comparison with other AR-enhanced models demonstrates that AR-ClinAgent surpasses state-of-the-art systems on all the major indicators, demonstrating the synergistic power of predictive intelligence, structured multi-agent coordination and procedural grounding. These findings demonstrate that AR-ClinAgent is not only helpful in assisting clinicians to follow the prescribed procedure accurately but also in improving the resource efficiency, so that it can be scaled up to use cases of simulation labs, manikin-based training and real-world clinician environment.

VI Conclusion

The AR-ClinAgent framework coupled with a set of PRA, MAO and RAG-based procedural grounding shows significant gains in the efficiency, accuracy and safety of clinical workflow. Baseline studies demonstrate that, compared with non-predictive AR systems, clinical

oversight interventions can be reduced by 31%, step-wise execution accuracy improved by 26% and near-misses cut by 22% using AR-ClinAgent. Moreover, 18% of instruction accessed latency can be saved and device level energy consumption is improved with 14% through task-distributed processing to increase not only the efficiency of work in operation but also that resources are utilized. The intuitive interface of the system supports several procedural tasks and displays depth-aware AR overlays with occlusion to a minimal extent such that it would be readable for clinicians. The multi-agent coordination enables the splitting of cognitive load between sub-agents specialised in, e.g., step prognostics, error verification and outcome validation. Procedural grounding with the RAG-based clinician-validated knowledge aligns the system well with real clinical practices, and thus enhances the decision quality. The evaluation experiment of simulation and manikin scenarios indicates the repeatability, security, as well as scalability of AR-ClinAgent. Its novel anticipated insight with semi-automatic AR guidance leads to addressing important limitations among existing AR-based clinical tools. For the next stage development, expansion of datasets diversity, integration with the patient-specific physiology modelling and extension to real-world clinical validation are means for enhancing applicability and generalisation potential of AR-ClinAgent under a wide variety of healthcare situations.

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